



ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS (SCI)

Bachelor of Data Science

COURSE NAME:

**PROBLEM SOLVING AND PROGRAMMING AND
PROGRAMMING FUNDAMENTALS**

INDIVIDUAL ASSIGNMENT

COURSE CODE: CCC1123

LECTURER NAME:

DR. BAHARUDDIN OSMAN & MR. ABU BAKAR NGAH

No.	Name	Student ID	Group
2.	Ismail Hosen	AIU25102207	C2 (BDS1A)

Program Summary

This program implements a Student Academic Performance System using the C programming language. It is designed to process and evaluate the academic results of five students.

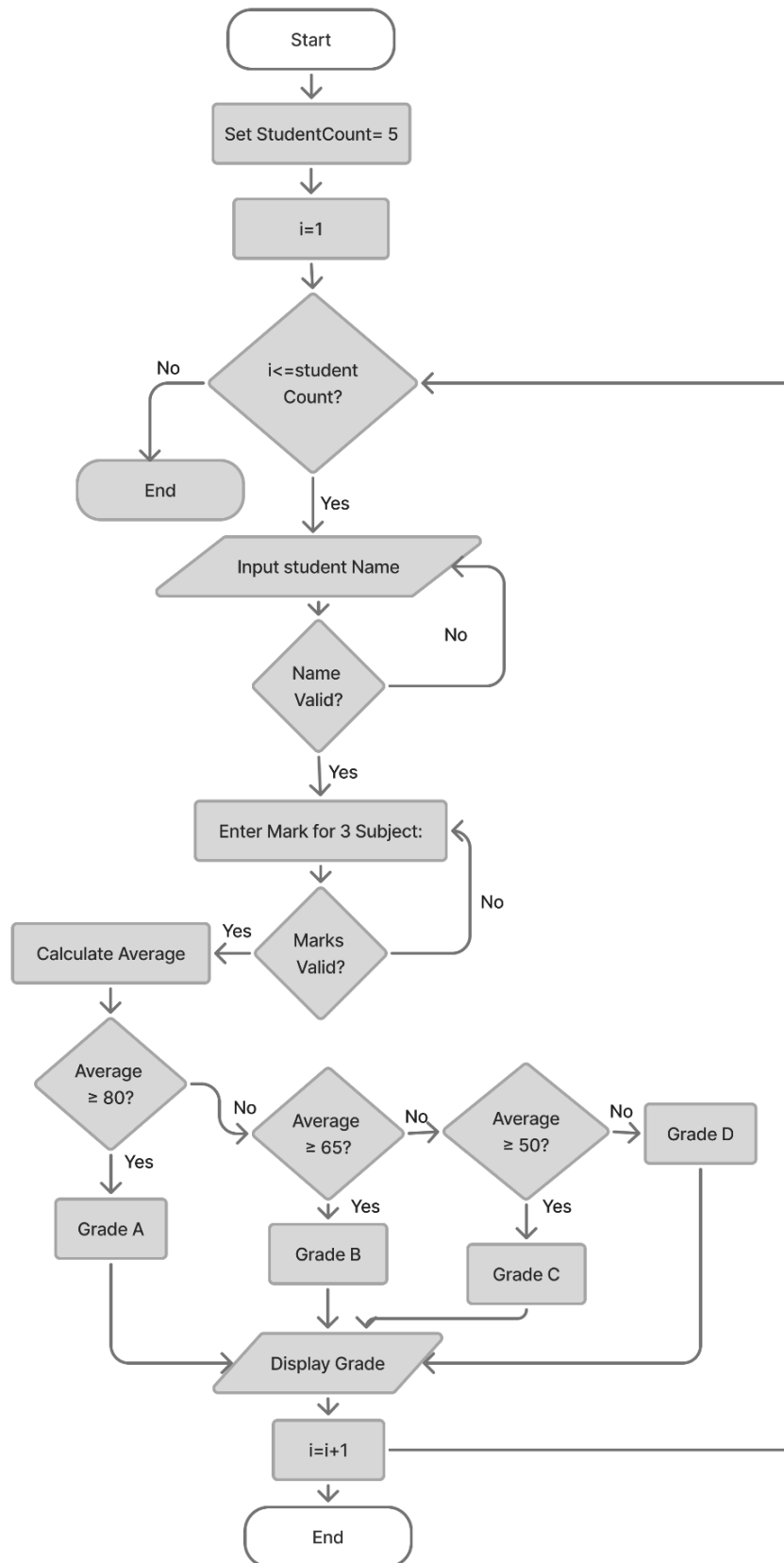
For each student, the program:

- Asks for the student's name and checks that it contains only letters and spaces.
- Takes marks for three subjects (Mathematics, Problem Solving, and Computer Architecture) and ensures the marks are between 0 and 100.
- Calculates the average mark using a user-defined function.
- Determines the student's grade using if-else statements conditional.
- Displays the student's name, average mark, and grade in a clear format.

The program uses:

- Functions to make the code organized and easy to understand.
- Loops to repeat the process for multiple students.
- Global and local variables to demonstrate variable scope.
- Input validation to avoid incorrect data entry.

Flow Chart



[Source code](#)

```
#include <stdio.h>

/*
    Global variable
    Stores total number of students
*/
int studentCount = 5;

/*
    Function: isValidName
    Purpose : Check whether the name contains
        - only letters and spaces
        - at least one alphabet letter
*/
int isValidName(char name[])
{
    int i;
    int hasLetter = 0; // To ensure name is not empty or spaces only

    for (i = 0; name[i] != '\0'; i++)
    {
        // Ignore newline character from fgets
        if (name[i] == '\n')
            continue;

        // Check if character is an alphabet letter
        if ((name[i] >= 'A' && name[i] <= 'Z') ||
            (name[i] >= 'a' && name[i] <= 'z'))
        {
            hasLetter = 1;
        }
    }
    // Allow space between names
```

```

        else if (name[i] == ' ')
        {
            continue;
        }
        // Any other character is invalid
        else
        {
            return 0;
        }
    }

    // Name is valid only if it contains at least one letter
    return hasLetter;
}

/*
Function: getValidName
Purpose : Repeatedly ask user to enter name
          until a valid name is provided
*/
void getValidName(char name[])
{
    while (1)
    {
        printf("Enter student name: ");
        fgets(name, 50, stdin);

        // Validate name
        if (isValidName(name))
        {
            // Remove newline character
            for (int i = 0; name[i] != '\0'; i++)
            {
                if (name[i] == '\n')
                {

```

```

        name[i] = '\0';
        break;
    }
}
return; // Exit function if name is valid
}
else
{
    printf("Invalid name! Use letters and spaces only.\n");
}
}
}

```

/*

Function: getValidMark

Purpose : Accept only numeric marks

Range must be between 0 and 100

*/

```
float getValidMark(char subject[])
```

```
{
```

```
    float mark;
```

```
    char input[50];
```

```
    char extra;
```

```
    while (1)
```

```
    {
```

```
        printf("Enter %s mark: ", subject);
```

```
        fgets(input, sizeof(input), stdin);
```

```
        // Check if input is a valid number only
```

```
        if (sscanf(input, "%f %c", &mark, &extra) != 1)
```

```
        {
```

```
            printf("Invalid input! Please enter numbers only.\n");
```

```
            continue;
```

```
        }
```

```

        // Check mark range
        if (mark < 0 || mark > 100)
        {
            printf("Mark must be between 0 and 100.\n");
            continue;
        }

        return mark; // Return valid mark
    }
}

/*
    Function: calculateAverage
    Purpose : Calculate average of three subjects
*/
float calculateAverage(float math, float ps, float ca)
{
    return (math + ps + ca) / 3;
}

/*
    Function: displayResult
    Purpose : Display student name, average and grade
*/
void displayResult(char name[], float avg, char grade)
{
    printf("\nStudent Name : %s", name);
    printf("\nAverage Mark : %.2f", avg);
    printf("\nGrade      : %c\n", grade);
}

/*
    Main function
    Controls the overall program execution

```

```

*/
int main()
{
    char name[50]; // Store student name
    float math, problemSolving, computerArchitecture;
    float average; // Store average mark
    char grade;    // Store grade

    // Loop to process all students
    for (int i = 1; i <= studentCount; i++)
    {
        printf("\n--- Student %d ---\n", i);

        // Get validated student name
        getValidName(name);

        // Get validated subject marks
        math = getValidMark("Mathematics");
        problemSolving = getValidMark("Problem Solving");
        computerArchitecture = getValidMark("Computer Architecture");

        // Calculate average marks
        average = calculateAverage(math, problemSolving, computerArchitecture);

        // Determine grade using if-else
        if (average >= 80)
            grade = 'A';
        else if (average >= 65)
            grade = 'B';
        else if (average >= 50)
            grade = 'C';
        else
            grade = 'F';

        // Display final result

```



```
        displayResult(name, average, grade);  
    }  
  
    return 0; // End of program  
}
```

Output:

Output

```
--- Student 1 ---
Enter student name: Ismail Hosen
Enter Mathematics mark: 46
Enter Problem Solving mark: 677
Mark must be between 0 and 100.
Enter Problem Solving mark: 78
Enter Computer Architecture mark: 78

Student Name : Ismail Hosen
Average Mark : 67.33
Grade       : B

--- Student 2 ---
Enter student name: nahid
Enter Mathematics mark: 34
Enter Problem Solving mark: 56
Enter Computer Architecture mark: 8

Student Name : nahid
Average Mark : 32.67
Grade       : F

--- Student 3 ---
Enter student name: Khan
Enter Mathematics mark: 23
Enter Problem Solving mark: 68
Enter Computer Architecture mark: 99

Student Name : Khan
Average Mark : 63.33
Grade       : C
```

```
Student Name : Khan  
Average Mark : 63.33  
Grade       : C
```

```
--- Student 4 ---
```

```
Enter student name: aMD  
Enter Mathematics mark: 88  
Enter Problem Solving mark: 78  
Enter Computer Architecture mark: 99
```

```
Student Name : aMD  
Average Mark : 88.33  
Grade       : A
```

```
--- Student 5 ---
```

```
Enter student name: aLI  
Enter Mathematics mark: 56  
Enter Problem Solving mark: 67  
Enter Computer Architecture mark: 89
```

```
Student Name : aLI  
Average Mark : 70.67  
Grade       : B
```

```
=== Code Execution Successful ===
```